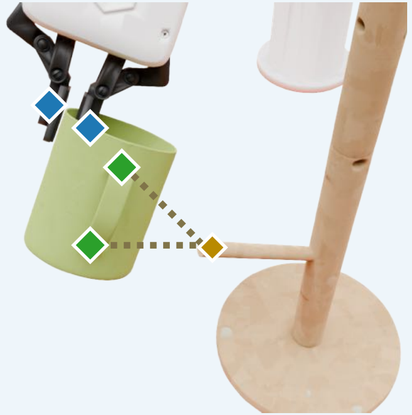


3D Keypoints



Motion Planning

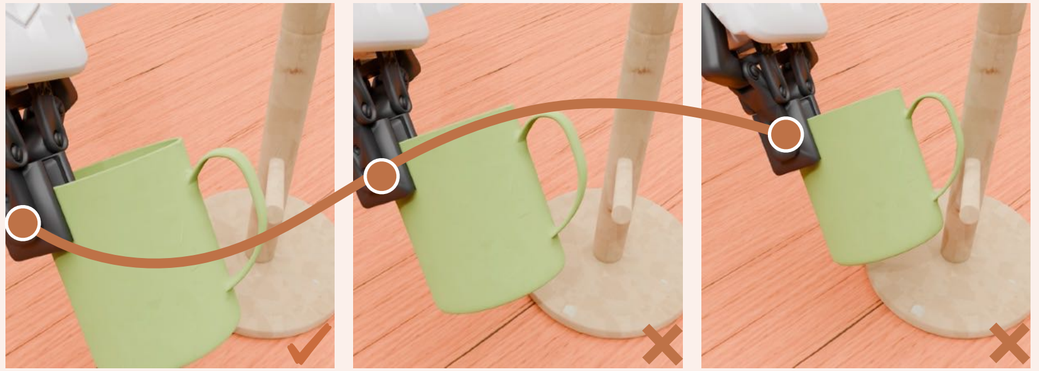
(a) Keypoint Programs

Keypoint Monitor

```
✓ # Stage1:
  def Grasp_Mug
    # … ◆ ◆
    Return EE_Diff

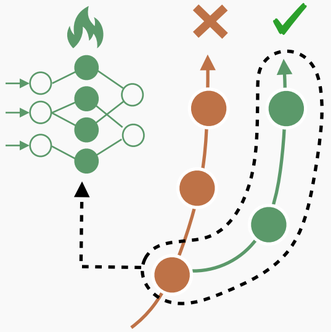
✗ # Stage2:
  def Hang_to_Rack
    # … ◆ ◆
    Return Object_Diff
```

Policy Rollout



(b) Automatic Failure Detection

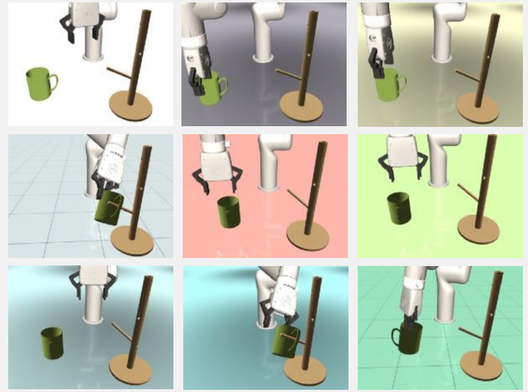
Policy Finetuning



Planner Correction



(c) Automatic Failure Correction



Domain Randomization



Real-World Rollout

(d) Zero-Shot Sim-to-Real